It appears as if the number of indecomposable representations does not depend on the orientation of the edges — and indeed Gabriel's theorem will generalize this observation.

6.4. Roots

From now on, let Γ be a fixed graph of type A_n, D_n, E_6, E_7, E_8. We denote the adjacency matrix of Γ by R.

Definition 6.4.1 (Cartan matrix)**.** We define the **Cartan matrix** of Γ as

$$A = 2\,\mathrm{Id} - R.$$

On the lattice $\mathbb{Z}^n$ (or the space $\mathbb{R}^n$) we then define an inner product

$$B(x, y) = x^T A y$$

corresponding to the graph Γ.

Lemma 6.4.2. (1) *B is positive definite.*

(2) *$B(x, x)$ takes only even values for $x \in \mathbb{Z}^n$.*

Proof. (1) This follows by definition, since Γ is a Dynkin diagram.

(2) By the definition of the Cartan matrix we get

$$\begin{aligned} B(x,x) = x^T A x &= \sum_{i,j} x_i\, a_{ij}\, x_j = 2\sum_i x_i^2 + \sum_{i,j,\, i\neq j} x_i\, a_{ij}\, x_j \\ &= 2\sum_i x_i^2 + 2\cdot \sum_{i<j} a_{ij}\, x_i x_j, \end{aligned}$$

which is even. □

Definition 6.4.3. A **root** with respect to a certain positive inner product is a shortest (with respect to this inner product) nonzero vector in $\mathbb{Z}^n$.

So for the inner product B, a root is a nonzero vector $x \in \mathbb{Z}^n$ such that

$$B(x, x) = 2.$$

Remark 6.4.4. There can be only finitely many roots, since all of them have to lie in some ball.